

Total No. of Questions : 8]

SEAT No. :

PA-962

[Total No. of Pages : 3

[5927]-414

**B.E. (Mechanical Engineering)
TURBOMACHINERY
(2019 Pattern) (Semester - VII) (402043)**

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Use of logarithmic tables, slide rule, and electronic pocket calculator is allowed.
- 4) Figure to the right indicate full marks.
- 5) Assume Suitable data, if necessary.

- Q1)** a) Classify Turbo machines. [2]
b) State Impulse Momentum Principle. Derive an expression for force exerted by jet of water on flat inclined plate. [6]
c) A Pelton wheel is to be designed for the following specifications: Shaft Power = 11,772kW; Head = 380 meters; Speed = 750 r.p.m; Overall efficiency = 86%; Jet diameter is not to exceed one sixth of the wheel diameter. Determine: [6]
i) The wheel Diameter
ii) The number of jets required, and
iii) Diameter of the jet.
Take $K_{v_1} = 0.985$ and $K_{u_1} = 0.45$

OR

- Q2)** a) Explain construction and working of Kaplan Turbine with neat sketch. [6]
b) What is the need of draft tube? Enlist types of draft tube with neat diagram. [4]
c) The external and internal diameters of inward flow reaction turbines are 1.20 m and 0.6 m respectively. The head on the turbine is 22 m and velocity of flow through the runner is constant and equal to 2.5 m/s. The guide blade is given as 10° and runner vanes are radial at inlet. If the discharge at outlet is radial, determine: [4]
i) The speed of the turbine
ii) The vane angle at the outlet of the runner

P.T.O.

- Q3) a)** What do you mean by compounding of steam turbines? Explain any one suitable example with neat sketch. [6]
- b)** In a De Laval Steam turbine steam issues from the nozzle with a velocity of 1200 m/s. The nozzle angle is 20° , the mean blade velocity is 400 m/s, the inlet and outlet angles of blades are equal. The mass of steam flowing through turbine per hour is 1000 kg. Determine the following by graphical method [6]
- i) Tangential Force
 - ii) Power Developed
 - iii) Blade Efficiency
- Take Velocity blade coefficient as 0.8

OR

- Q4) a)** Explain governing of steam turbine with any one method. [6]
- b)** Following data refers to particular stage of Parson's reaction turbine. [6]
The speed of turbine = 1500 rpm
Mean Diameter of rotor = 1 m
Stage efficiency = 0.8
Blade outlet angle = 20°
Speed Ratio = 0.7
Determine the available enthalpy drop in the stage by graphical method.
- Q5) a)** State & explain: [6]
- i) Unit Speed
 - ii) Unit Discharge
 - iii) Unit Power
- b)** A centrifugal pump delivers water against a net head of 14.5 meters and a design speed of 1000 r.p.m. The vanes are curved back to an angle of 30° with the periphery. The impeller diameter is 300 mm and outlet width is 50 mm. Determine the discharge of the pump if manometric efficiency is 95%. [6]

OR

- Q6) a)** Explain the following: [6]
- Priming of Centrifugal Pump
 - Net Positive Suction Head
 - Cavitation
- b)** A centrifugal pump discharges $0.15 \text{ m}^3/\text{s}$ of water against head of 12.5 m , the speed of impeller being 600 r.p.m. The outer and inner diameters of impeller are 500 mm and 250 mm respectively and the vanes are bent back at 35° to the tangent at exit. If the area of flow remains 0.07 m^2 from inlet to outlet, Calculate: [6]
- Manometric Efficiency of pump
 - Vane angle at inlet
- Q7) a)** Explain construction and working of Axial flow Compressor with neat diagram. [6]
- b)** A centrifugal compressor develops a pressure ratio of 5 and an air consumption of 30 kg/s . The inlet temperature and pressure are 15°C and 1 bar respectively. If isentropic efficiency is 0.85, Calculate [6]
- Work done
 - Exit total temperature
 - The power required

OR

- Q8) a)** Explain surging and choking phenomenon in centrifugal compressors with neat diagram. [6]
- b)** Differentiate between centrifugal compressor and axial flow compressor. [4]
- c)** Give classification of centrifugal compressors. [2]
